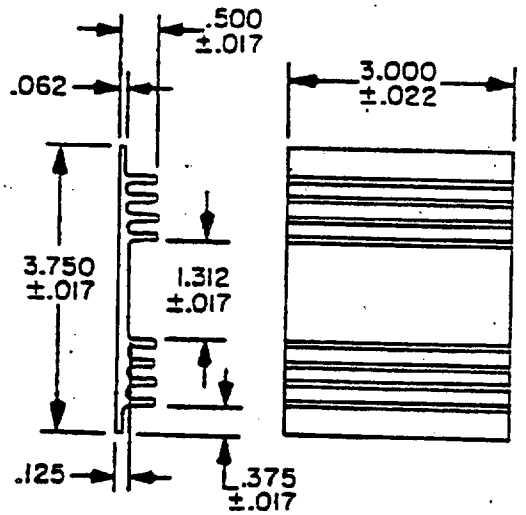
MILITARY SPECIFICATION SHEET

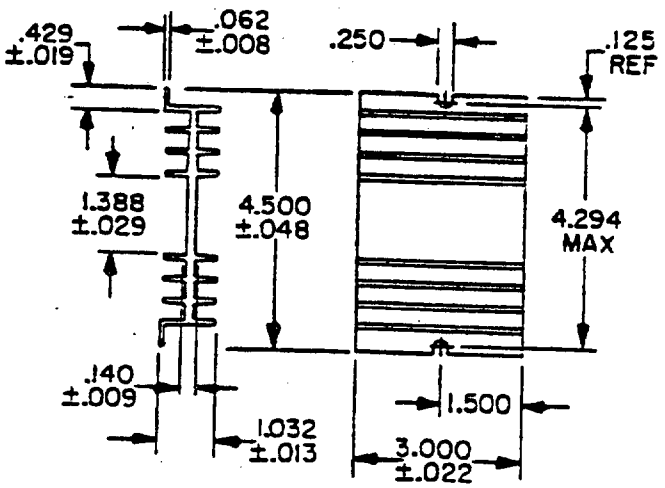
HEAT SINKS, SEMICONDUCTOR DEVICES, CLASS 1 (EXTRUDED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the heat sinks described herein shall consist of this specification and the latest issue of MIL-H-87111.



CONFIGURATION A

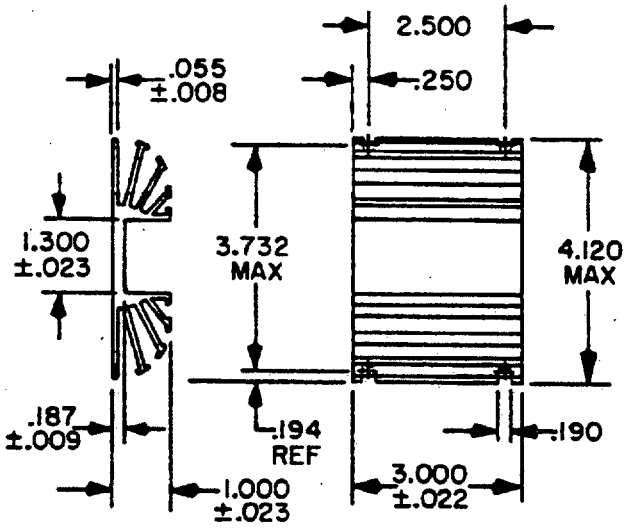


CONFIGURATION B

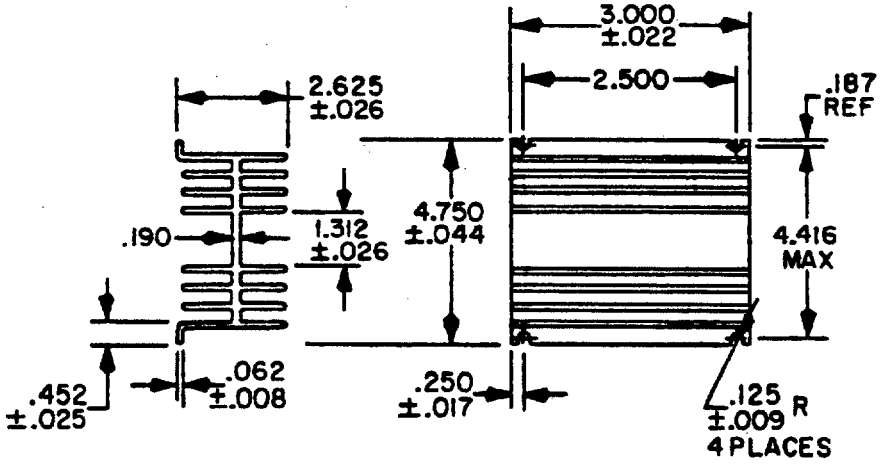
FIGURE 1. Heat sinks (extruded).

Ⓑ denotes changes

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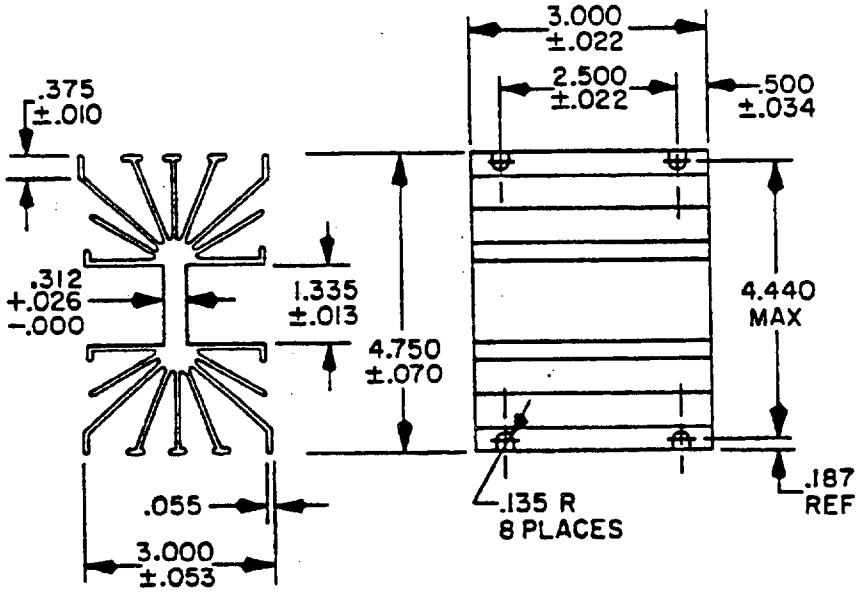
CONFIGURATION C



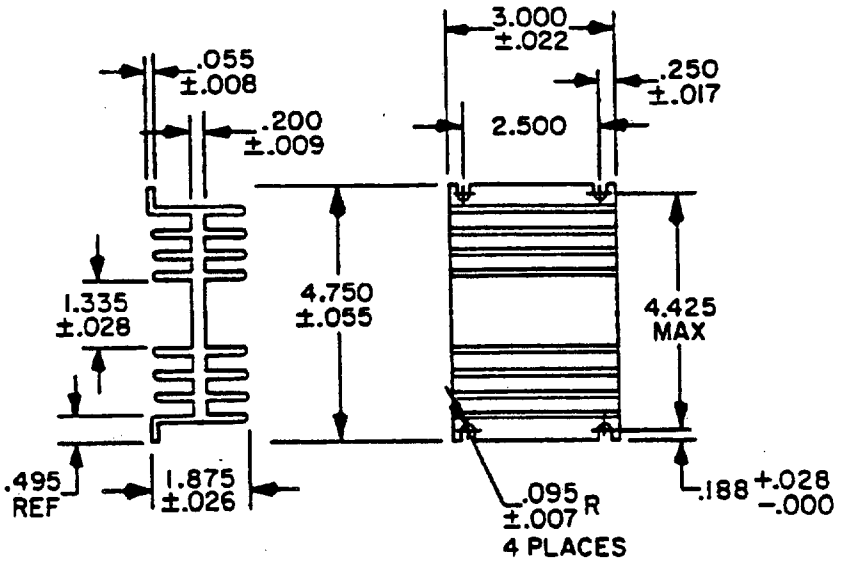
CONFIGURATION D

FIGURE 1. Heat sinks (extruded) - Continued.

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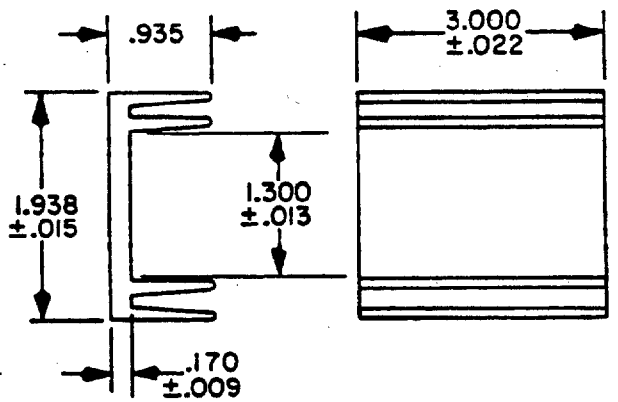
CONFIGURATION E



CONFIGURATION F

FIGURE 1. Heat sinks (extruded) - Continued.

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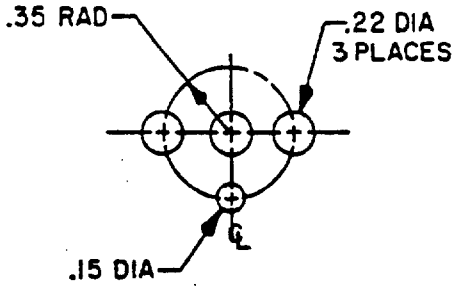
CONFIGURATION G

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.008	0.20	.026	0.66	.095	2.41	.250	6.35	1.300	33.02	3.732	94.79
.009	0.23	.028	0.71	.125	3.18	.312	7.92	1.312	33.32	3.750	95.25
.010	0.25	.029	0.74	.135	3.43	.375	9.52	1.335	33.91	4.120	104.65
.013	0.33	.034	0.86	.140	3.56	.429	10.90	1.388	35.26	4.294	109.07
.015	0.38	.044	1.12	.170	4.32	.452	11.48	1.500	38.10	4.416	112.17
.017	0.43	.048	1.22	.187	4.75	.495	12.57	1.875	47.62	4.425	112.40
.019	0.48	.053	1.35	.188	4.78	.500	12.70	1.938	49.23	4.440	112.78
.022	0.56	.055	1.40	.190	4.83	.935	23.75	2.500	63.50	4.500	114.30
.023	0.58	.062	1.57	.194	4.93	1.000	25.40	2.625	66.68	4.750	120.65
.025	0.64	.070	1.78	.200	5.08	1.032	26.21	3.000	76.20		

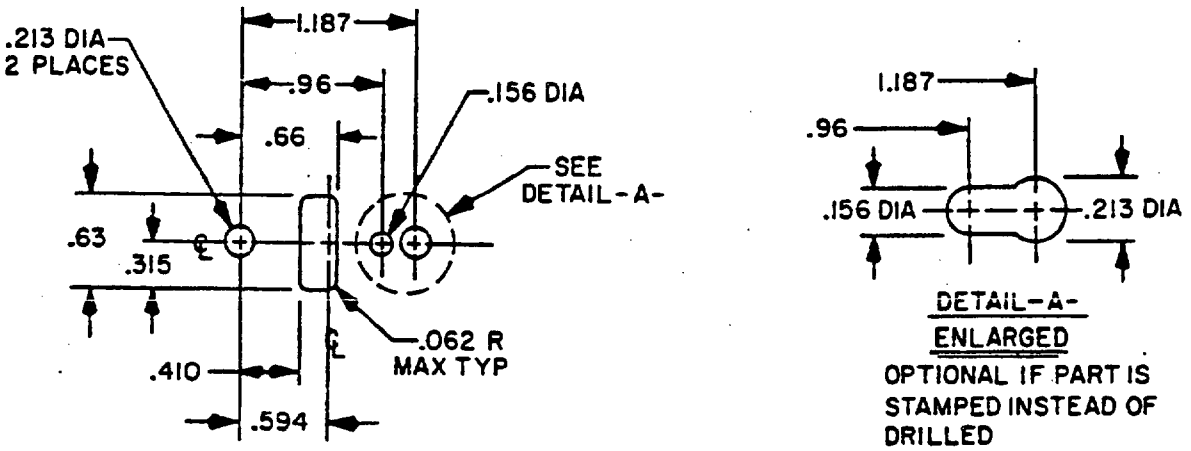
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.011 (0.28 mm) for decimals. Cut-off tolerance shall be ±.022 (0.56 mm) unless otherwise specified.

FIGURE 1. Heat sinks (extruded) - Continued.

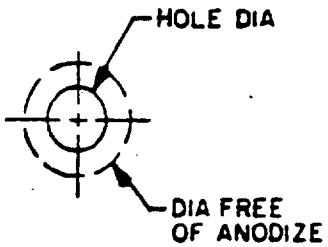
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H1 (TO-36 JEDEC case, TO-6 JEDEC case)



H2 (Universal, TO-3, TO-66, TO-126, and TO-220)

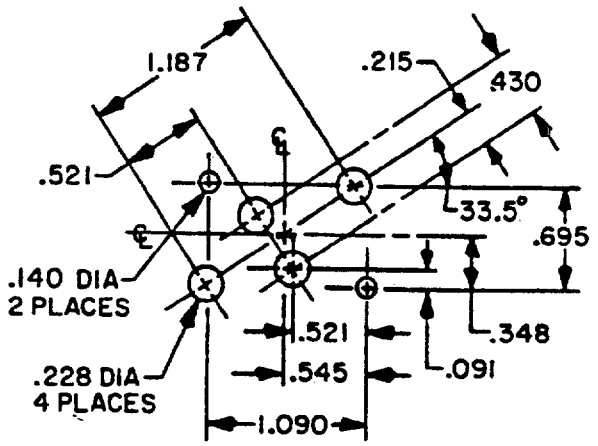


Hole diameter	Diameter free of anodize ±.035 (0.89)	Hole pattern number
.200 ± .005 (5.08) (0.13)	.500 (12.70)	H3
.270 ± .007 (6.86) (0.18)	.750 (19.05)	H4
.330 ± .007 (8.38) (0.18)	1.000 (25.40)	H5

H3, H4 and H5 (stud)

FIGURE 2. Mounting hole patterns.

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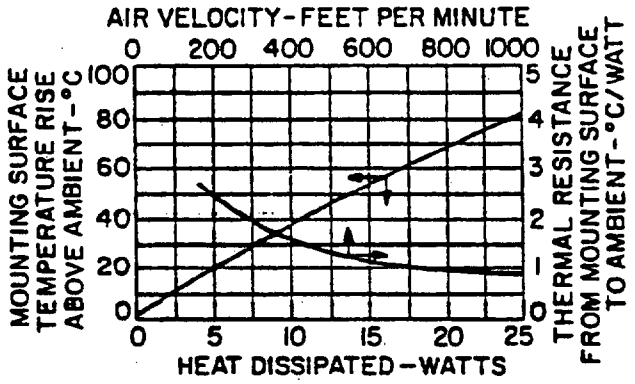
H6 (TO-3 socket JEDEC case)

INCHES	MM
.062	1.57
.091	2.31
.098	2.49
.140	3.56
.15	3.8
.152	3.86
.156	3.96
.213	5.41
.215	5.46
.22	5.6
.228	5.79
.315	8.00
.348	8.84
.35	8.9
.410	10.41
.430	10.92
.500	12.70
.521	13.23
.545	13.84
.593	15.06
.594	15.09
.63	16.0
.66	16.8
.695	17.65
.96	24.4
1.090	27.69
1.187	30.15

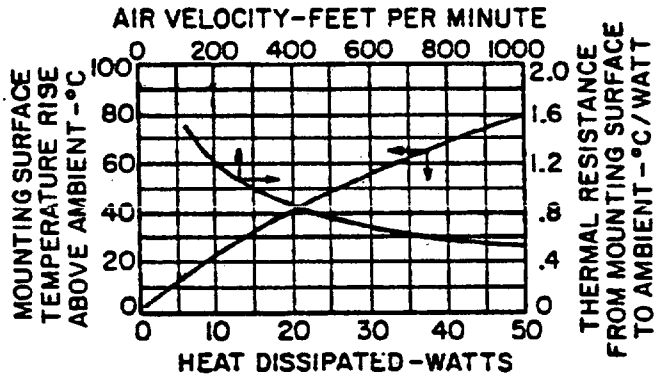
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm) on decimals and ±2° on angles. A .062 (1.57 mm) radius is permissible on corners.
4. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes.

FIGURE 2. Mounting hole patterns - Continued.

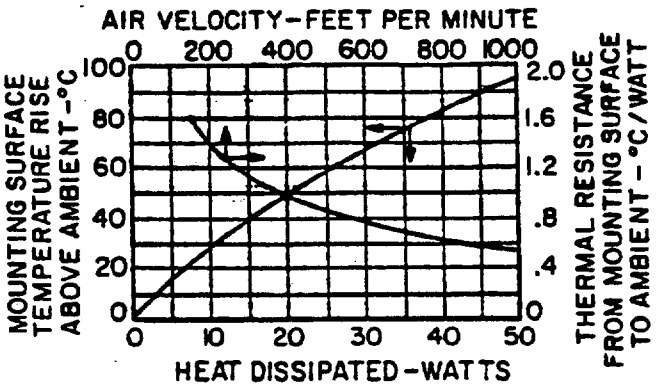
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CONFIGURATION A



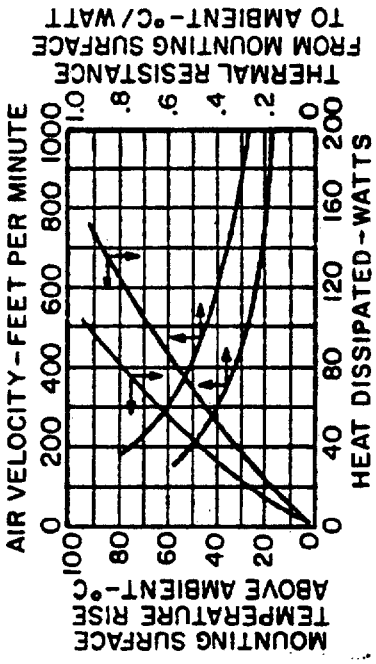
CONFIGURATION B



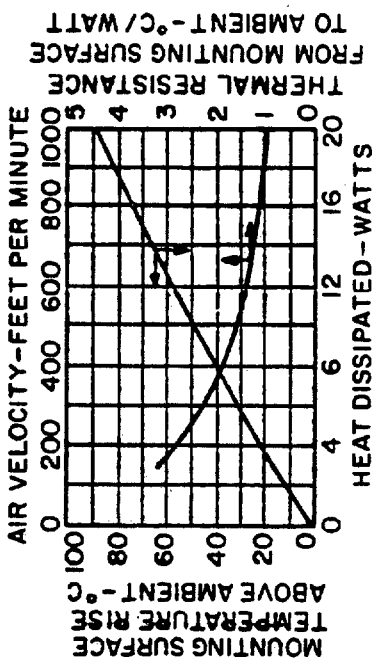
CONFIGURATION C

FIGURE 3. Thermal performance characteristic.

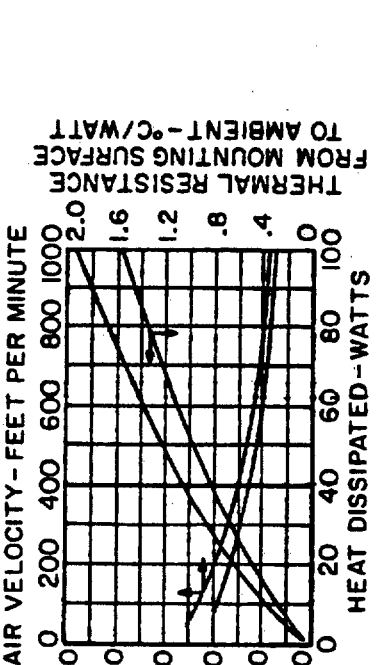
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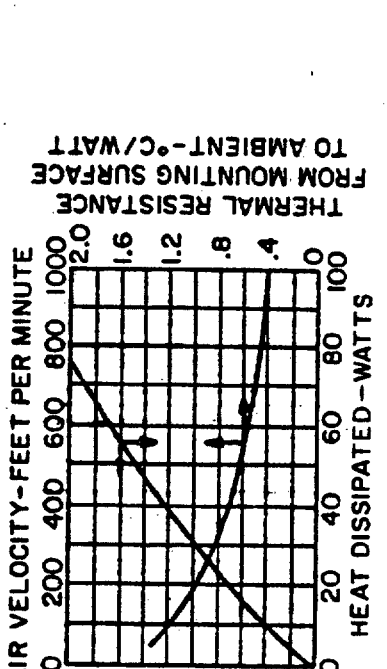
CONFIGURATION E



CONFIGURATION G



CONFIGURATION D



CONFIGURATION F

FIGURE 3. Thermal performance characteristic - Continued.

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REQUIREMENTS:

Design and configuration: See figures 1 and 2.

Material: Aluminum alloy (class 1).

Finish: Black anodize, insulating coating, hard anodize, or gold chromate. (Finish A, C, H, or R.)

Thermal resistance: 4°C/watt maximum at any power dissipation below 20 watts. 3°C/watt maximum at 20 watts. Based upon 3 inch length of extrusion.

Mounting surface flatness: The maximum allowable deviation from flat on semiconductor mounting surfaces shall not exceed .004 inch per inch TIR.

Surface roughness: Mounting surface roughness shall not exceed 63 microinches rms.

JEDEC case size: See figure 2.

Part number: M87111/06- (dash number from table I).

NOTE: The recommended thermal performance characteristics in forced air and natural convection, as opposed to power dissipation, is shown on figure 3.

Ⓑ TABLE I. Part numbers.

Part number M87111/06- 1/						Configuration
Hole pattern (see figure 2)						
H1	H2	H3	H4	H5	H6	
1-01	1-08	1-15	1-22	1-29	1-36	A
1-02	1-09	1-16	1-23	1-30	1-37	B
1-03	1-10	1-17	1-24	1-31	1-38	C
1-04	1-11	1-18	1-25	1-32	1-39	D
1-05	1-12	1-19	1-26	1-33	1-40	E
1-06	1-13	1-20	1-27	1-34	1-41	F
1-07	1-14	1-21	1-28	1-35	1-42	G

1/ Complete dash number will include an additional letter to indicate finish (i.e.,A, C, H, or R). Example: M87111/06-1A01.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - TD
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-87111/6B	HEAT SINKS, SEMICONDUCTOR DEVICE, CLASS 2 AND 3 (RETAIN CLIP TYPE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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